Tahdhīb al-Manṭiq

تهذيب المنطق

Saʿd al-Dīn al-Taftāzānī · Continuous bilingual 19-unit logic working text

> **Textual qualification.** Nineteen of nineteen units translated; detailed word-by-word 1912 print and manuscript verification outstanding. Modal logic and syllogistic validity require specialist audit.

Editorial division 1

L.01

Opening dedication and preface

Praise belongs to God, who has guided us to the straight course and made success [tawfīq] our best companion. Blessings and peace be upon the one whom He sent as a guidance worthy of being followed in finding guidance, and as a light whose example deserves to be followed; and upon his family and companions, who attained felicity, on the paths of truthfulness, through assent [taṣdīq], and ascended, on the steps of truth, through verification [taḥqīq]. To proceed: this is the utmost refinement of discourse, in setting forth logic and kalām, and in bringing near the intended aim through setting out the doctrines of Islam. I have made it an illumination for whoever seeks insight in the course of instruction, and a reminder for whoever among those possessing understanding wishes to recollect; and especially for the dearest, most affectionate son deserving of honour, the namesake of the Beloved of God—upon him greetings and peace. May success always sustain him, and divine support protect him. Our reliance is upon God, and in Him we seek protection.

Editorial division 2

L.02

Knowledge, rational inquiry and the subject of logic

If knowledge is assent to a relation, it is judgment (taṣdīq); otherwise, conception (taṣawwur). Both are divided into what is immediate and what is acquired through rational consideration. Rational consideration is attention to the intelligible in order to attain what is unknown. Error may occur in it, and so a rule was needed to protect the mind against error: logic. Its subject is what is already known, whether conceptual or propositional, insofar as this leads to a sought conception, called a means of definition (muʿarrif), or to a sought judgment, called a proof (ḥujja).

Editorial division 3

L.03

Signification, simple and compound expressions, and kinds of names

Signification of the whole instituted meaning is correspondence (muṭābaqa), of part of that meaning inclusion (taḍammun), and of something outside it implication (iltizām). Implication requires a rational or customary necessary connection; the latter two kinds of signification presuppose correspondence, at least implicitly, though not the converse. An expression is compound if its constituent parts are intended to signify constituents of its meaning: it may be complete, either an assertive statement or a nonassertive utterance, or incomplete, whether restrictive or otherwise. If not, it is simple. If it signifies independently with reference to one of the three times, it is a verb (kalima); without that temporal feature, a noun; without independent signification, a particle. When an expression has one meaning instituted for a particular person or thing it is a proper name; otherwise, it is univocal when its instances stand equally under that meaning, and analogically graded (mushakkik) when they differ by priority or precedence. If several meanings are independently instituted for it, it is equivocal (mushtarak). If a second meaning becomes established it is a transferred term attributed to the person or convention responsible for the transfer; otherwise, we distinguish literal and figurative usage.

Editorial division 4

L.04

Particular and universal concepts; degrees of instantiation

Chapter. If it is impossible to suppose that a concept truly applies to many [individuals], it is a particular [juzʾī]; otherwise it is a universal [kullī]. Instances of a universal may be impossible; or they may be possible but not exist; or only one may exist, either with others possible or with others impossible; or many may exist, whether finite or infinite in number.

Editorial division 5

L.05

Relations between two universals

If two universals are entirely separate in extension, they are disjoint. Otherwise, if they coincide in their instances universally in both directions, they are coextensive, and their contradictories are likewise coextensive. If the inclusion is universal in only one direction, one is absolutely more general and the other absolutely more specific, while the relation between their contradictories is reversed. Otherwise they overlap in some respects; their contradictories exhibit partial exclusion, as do those of disjoint universals. The term ‘particular’ may also designate what is more specific relative to another, and in this relative sense it extends beyond the absolute particular.

Editorial division 6

L.06

The five predicables: genus

The five predicables are as follows. First, genus: what is predicated of many things differing in their specific natures in answer to ‘What is it?’ If the answer, for a quiddity and some of the things sharing its genus, is also the answer for all such sharers, it is a proximate genus, such as ‘animal’; otherwise a remote genus, such as ‘growing body’.

Editorial division 7

L.07

Species, differentia, property and common accident

Second, species: what is predicated of many things agreeing in specific nature in answer to ‘What is it?’ The word also applies to a quiddity of which, and of something else, a genus is predicated; the latter is a relative species, the former a real species. They overlap in human nature, while ‘animal’ and ‘point’ distinguish their applications. Genera can ascend toward a highest genus such as substance; species descend toward a lowest species, with intermediate ranks between. Third, differentia: what is predicated in answer to ‘What sort of thing is it in itself?’ It is proximate when it distinguishes something from those sharing its proximate genus, otherwise remote. Relative to what it constitutes it is constitutive, and relative to what it divides from it is divisive. What constitutes a higher kind also constitutes its subordinate kind, but not conversely; for what divides, the direction is reversed. Fourth, property: an extrinsic predicable belonging exclusively under one specific nature. Fifth, common accident: an extrinsic predicable applicable also to other natures. Either may be inseparable from its subject, in which case it is a concomitant, considered through essence or existence. An evident concomitant is known as following from the conception of the subject alone or from conceiving subject and concomitant together; otherwise it is non-evident. What may separate is a separable accident, whether enduring or departing quickly or slowly.

Editorial division 8

L.08

Three senses of ‘universal’

Conclusion. The concept of universality is the logical universal; the nature of which universality is predicated is the natural universal; and the two considered together are the intellectual universal. The same distinction applies to the five predicables. The correct account of the natural universal’s existence is that its individual instances exist.

Editorial division 9

L.09

Definitional accounts

Chapter on accounts that make things known and their divisions. A defining account is what is said to furnish conception of the thing; it must be coextensive with its object and clearer. Hence one cannot properly define through what is more general or more specific, equally known or unknown, or less intelligible. Explanation by proximate differentia constitutes an essential definition (ḥadd), and by a property a descriptive account (rasm). Either is complete when joined to the proximate genus; otherwise it is incomplete. The common accident is not counted as the required element. An incomplete account is sometimes permitted to be more general, as with a verbal explanation clarifying the meaning signified by an expression.

Editorial division 10

L.10

Judgments: categorical propositions, quantification, existential import

Chapter on judgments. A proposition [qaḍiyya] is a discourse capable of truth or falsity. If it judges that something belongs to something else or is denied of it, the proposition is categorical [ḥamliyya], affirmative or negative. The thing judged about is the subject [mawḍūʿ], and what is affirmed or denied is the predicate [maḥmūl]. What indicates their predicative relation is the copula [rābiṭa], for which the pronoun huwa (“is/he”) has been adopted [as a linguistic sign]. Otherwise the proposition is conditional [sharṭiyya], whose first component is the antecedent [muqaddam] and the second the consequent [tālī]. If the subject is a determinate individual the categorical proposition is singular [shakhṣiyya] or specific [makhṣūṣa]. If it is the nature itself, the proposition is natural [ṭabīʿiyya]. Otherwise, if the number of its individuals is specified as all or some, it is quantified [maḥṣūra], universal or particular, and the expression making the quantity explicit is the quantifier [sūr]; otherwise it is indeterminate [muhmala], equivalent [in inferential effect] to a particular proposition. An affirmative [categorical proposition] requires the subject to exist: either as actually realised, in which case the proposition is external [khārijiyya]; or as supposed/estimated, in which case it is real or essentially quantified [ḥaqīqiyya]; or mentally, in which case it is mental [dhihniyya].

Editorial division 11

L.11

Infinite-negative and modal categorical propositions

A negation particle may be made part of one of the terms of a proposition, in which case it is an infinitated proposition (maʿdūla); otherwise it is non-infinitated (muḥaṣṣala). The manner of the predicative relation may be expressly stated, giving a modal proposition (muwajjaha); the expression stating that manner is its modality (jiha). Otherwise it is unqualified or absolute (muṭlaqa). If it judges the relation necessary for as long as the subject itself exists, it is absolutely necessary (ḍarūriyya muṭlaqa); if for as long as the subject bears its specified description, it is generally conditioned (mashrūṭa ʿāmma); if at a specified time, absolutely temporal (waqtiyya muṭlaqa); if at some unspecified time, absolutely diffuse (muntashira muṭlaqa). If it judges the relation to endure for as long as the subject exists, it is absolutely permanent (dāʾima muṭlaqa); if for as long as the subject bears its description, generally customary (ʿurfiyya ʿāmma). If it affirms the actuality of the relation, it is generally absolute (muṭlaqa ʿāmma); if it denies the necessity of the opposing relation, generally possible (mumkina ʿāmma). These are the simple modal forms. The two general forms—the conditioned and the customary—and the two temporal forms—the temporal and the diffuse—may be qualified by the nonpermanence of their attribution to the subject as such. They are then called, respectively, the specifically conditioned (mashrūṭa khāṣṣa), specifically customary (ʿurfiyya khāṣṣa), specifically temporal (waqtiyya), and specifically diffuse (muntashira). The generally absolute may be qualified by the absence of essential necessity, becoming existential nonnecessary (wujūdiyya lā ḍarūriyya), or by the absence of essential permanence, becoming existential nonpermanent (wujūdiyya lā dāʾima). The generally possible may be qualified by the absence of necessity on the corresponding side as well, becoming specifically possible (mumkina khāṣṣa). These are compound modalities: a nonpermanence qualification amounts to a generally absolute proposition of the opposite quality but the same quantity as that which it qualifies; a nonnecessity qualification similarly amounts to a generally possible proposition of opposite quality and corresponding quantity.

Editorial division 12

L.12

Conditional propositions

Chapter. A conditional proposition is connected [muttaṣila] if it judges that one relation holds, or does not hold, on the supposition of another. It is implicative [luzūmiyya] when there is a connection between the two relations; otherwise it is coincidental [ittifāqiyya]. It is disjunctive [munfaṣila] if it judges that the two relations are incompatible, or not incompatible, with respect to truth and falsity together, in which case it is the true disjunctive [ḥaqīqiyya]; or with respect to truth only, in which case it prevents conjunction [māniʿat al-jamʿ]; or with respect to falsity only, in which case it prevents joint exclusion [māniʿat al-khulū]. Each [of these disjunctive types] is essentially opposed [ʿināḍiyya] if the exclusion arises from the natures of its two components, and otherwise is coincidental. Again, if the judgment in a conditional holds under all suppositions concerning the antecedent, it is universal; if under some, without specification, particular; if under a determinate supposition, singular; otherwise indeterminate. The two members of a conditional are originally two categorical propositions, two connected conditionals, two disjunctives, or two different kinds. The addition of the connective or disjunctive particle deprives them of [their independent] completeness.

Editorial division 13

L.13

Contradictory propositions

Contradiction is the opposition of two propositions such that the truth of either requires the falsity of the other, and conversely. They differ in quantity, quality and modal qualification, but agree in the other respects. The contradictory of a necessary proposition is generally possible; of a permanent proposition, generally absolute; of a generally conditioned proposition, a temporally possible form; and of a generally customary proposition, a temporally absolute form. In the compound case the relevant sense alternates between the contradictories of its components, with the particular force assessed in relation to each individual.

Editorial division 14

L.14

Simple conversion (ʿaks mustawī)

Simple conversion exchanges a proposition’s two terms while preserving truth and quality. An affirmative converts only to a particular affirmative, since its predicate—or a conditional’s consequent—may be more general. A universal negative converts universally; otherwise something would have to be denied of itself. A particular negative does not convert, since its subject—or antecedent—may be general. With respect to modality, the two permanent and two general affirmative forms convert into a temporally qualified absolute; the two special forms into a temporally qualified nonpermanent; and the temporal, existential and general absolute forms into a general absolute. Neither possibility form converts. Among negatives, the two permanent forms convert as permanent, the two general forms as generally customary, and the two special forms, in some cases, as customary nonpermanent. Each valid conversion can be demonstrated by deriving an impossibility from the original and the contradictory of its converse; the invalid cases are defeated by counterexample.

Editorial division 15

L.15

Conversion by contraposition (ʿaks al-naqīḍ)

Chapter. Conversion by contraposition [ʿaks al-naqīḍ] consists in interchanging the contradictories of the two terms while retaining truth and quality; or in placing the contradictory of the second term first while reversing the quality [of the predication]. The rules for affirmative propositions here are those of negative propositions in simple conversion, and conversely. The proof is the same proof and the counterexample the same counterexample. By assuming an instance [bi-l-iftirāḍ], it has been shown that the two specific forms can be converted from the particular affirmative here, and from the particular negative there [under simple conversion], to the specific customary form. Reflect upon this.

Editorial division 16

L.16

Syllogism, terms, figures, and the four sets of conditions

Chapter. A syllogism [qiyās] is a discourse composed of propositions from which, by virtue of that composition itself, another proposition necessarily follows. If the resulting proposition is mentioned within it in matter and form [bi-māddatihi wa-hayʾatihi], the syllogism is exceptive [istisnāʾī]; otherwise it is conjunctive [iqtirānī], either categorical or conditional. In a categorical syllogism, the subject of the sought conclusion is called the minor term [al-aṣghar]; its predicate, the major term [al-akbar]; and what is repeated, the middle term [al-awsaṭ]. The premise containing the minor term is the minor premise [al-ṣughrā], and that containing the major term the major premise [al-kubrā]. The middle term may be the predicate of the minor and the subject of the major: this is the first figure. Or it may be the predicate of both: the second figure. Or it may be the subject of both: the third figure. Or the reverse of the first arrangement: the fourth figure. First figure. The minor must be affirmative and actual [fiʿliyya], together with a universal major. It then produces, with two affirmative premises and a universal affirmative major, an affirmative conclusion [universal or particular according to the quantity of the minor]; with a universal negative major, a negative conclusion [universal or particular according to the minor], necessarily. Second figure. The premises must differ in quality, and the major premise must be universal; together with permanence of the minor premise, or the possibility of converting the negative major, or the presence of a possible premise with a necessary one, or a conditional major. Two universal premises then yield a universal negative; and where their quantities also differ, a particular negative. This is shown by reductio [khalf], or by conversion of the major or the minor, followed by arranging [the converted premises] and deriving the conclusion. Third figure. The minor premise must be affirmative and actual, with at least one premise universal. Two affirmatives, with the major universal, or the reverse, yield a particular affirmative; and with a universal negative, or with a universal and a particular [of opposing qualities], they yield a particular negative. This is established by reductio, or by conversion of minor or major followed by arranging and deriving the conclusion. Fourth figure. The premises must both be affirmative with the minor universal, or must differ in quality with at least one universal. A universal affirmative joined with [any of] the four [appropriate forms], a particular affirmative with a universal negative, either of the two negative forms paired with a universal affirmative, and the universal [negative] with a particular affirmative yield a particular affirmative if the result is not obtained by negation, and otherwise a particular negative. Proof is by reductio, or by reversing the order and then converting the conclusion, or by converting both premises, or by reduction to the second figure through conversion of the minor, or to the third through conversion of the major.

Editorial division 17

L.17

Conjunctive conditional and exceptive syllogisms

Chapter. The conditional variety of conjunctive syllogism may be composed of two connected conditionals, two disjunctives, a categorical and a connected conditional, a categorical and a disjunctive, or a connected and a disjunctive proposition. The four figures can be formed in it. A detailed treatment would be lengthy. Chapter. The exceptive syllogism yields a conclusion, from a connected conditional, by affirming its antecedent or denying its consequent; from a true disjunctive, it yields a conclusion when either member is affirmed, as in the preventing-conjunction form, or denied, as in the preventing- exclusion form. It may specifically be called a syllogism by reductio [qiyās al-khalf]: this is the procedure that aims to establish what is sought by invalidating its contradictory. It reduces to a combination of an exceptive and a conjunctive syllogism.

Editorial division 18

L.18

Induction, analogy, and the five kinds of syllogistic discourse

Chapter. Induction [istiqrāʾ] is a survey of particulars in order to establish a universal judgment. Analogy [tamthīl] is the disclosure that one particular shares with another the cause of a judgment, in order to establish that judgment of the former. The principal basis of its method is concomitant variation [dawarān] and disjunctive elimination [tardīd]. Chapter. A syllogism is either demonstrative [burhānī], being composed of certainties [yaqīniyyāt]. Their fundamental kinds are first principles [awwaliyyāt], direct observations [mushāhadāt], experiential judgments [tajribiyyāt], intuitive judgments [ḥadsiyyāt], multiply transmitted judgments [mutawātirāt], and theoretical judgments [naẓariyyāt] (Cairo reading). If the middle term is a cause of the predicative relation not only in the mind but also in reality, the demonstration is demonstration of why [limmī]; otherwise it is demonstration of that [innī]. Or a syllogism is dialectical [jadalī], composed of generally accepted and conceded premises [mashhūrāt, musallamāt]; rhetorical [khiṭābī], composed of accepted on authority and probable premises [maqbūlāt, maẓnūnāt]; poetic [shiʿrī], composed of imaginative premises [mukhayyalāt]; or sophistical [safsaṭī], composed of estimative and deceptively similar premises [wahmiyyāt, mushabbahāt].

Editorial division 19

L.19

Conclusion: the constituents of a science (Cairo 1912)

The constituents of the sciences are three: subjects, principles and investigated problems. Subjects are those things whose essential properties the science investigates. Principles include the definitions of subjects, their parts and attributes, and self-evident or borrowed premises on which the science’s arguments are built. Problems are propositions investigated in the science: their subjects are the subject matter of the discipline itself, a species of it, one of its essential properties, or a compound of these; their predicates are matters external to those subjects yet attaching to them in virtue of their nature. The word ‘principles’ may also designate what is presented before the principal aim, and ‘preliminaries’ what one must know in order to undertake an inquiry competently. God knows best what is correct.